IT114 Project Milestone 1

Student submission

Assignment revision: 2 | Maximum points: 10 | Student: ja692 | Submitted: 2026-11-01T18:36:28.435Z | Submission revision: 303 | Receipt: v4r_01M6C6C4CMS178WSYKWGZN6SVF | Exported: 2026-11-01T18:36:30.588Z

Worksheet link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-project-milestone-1/ja692

Grading link

https://courses.ethereallab.app/assignment/v4/IT114-001-F2026/it114-project-milestone-1/grading/ja692

AI disclosure: Completed; 1 entry; 1 declared AI use; 0 declared no AI use; policy v1

Server startup

Demonstrate that the server starts from the repository root, listens, and accepts connections.

Server source

URL prompt

Provide direct GitHub file URLs for the server entry point and connection-accepting code on `Milestone1`; add multiple URLs if those are in separate files.

Requirements and potential deductions

- The linked source starts a listener and accepts incoming connections. (up to -0.55 points)
- Relevant code comments identify the student's UCID, date, and purpose. (up to -0.2 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Server/Server.java

Listening output

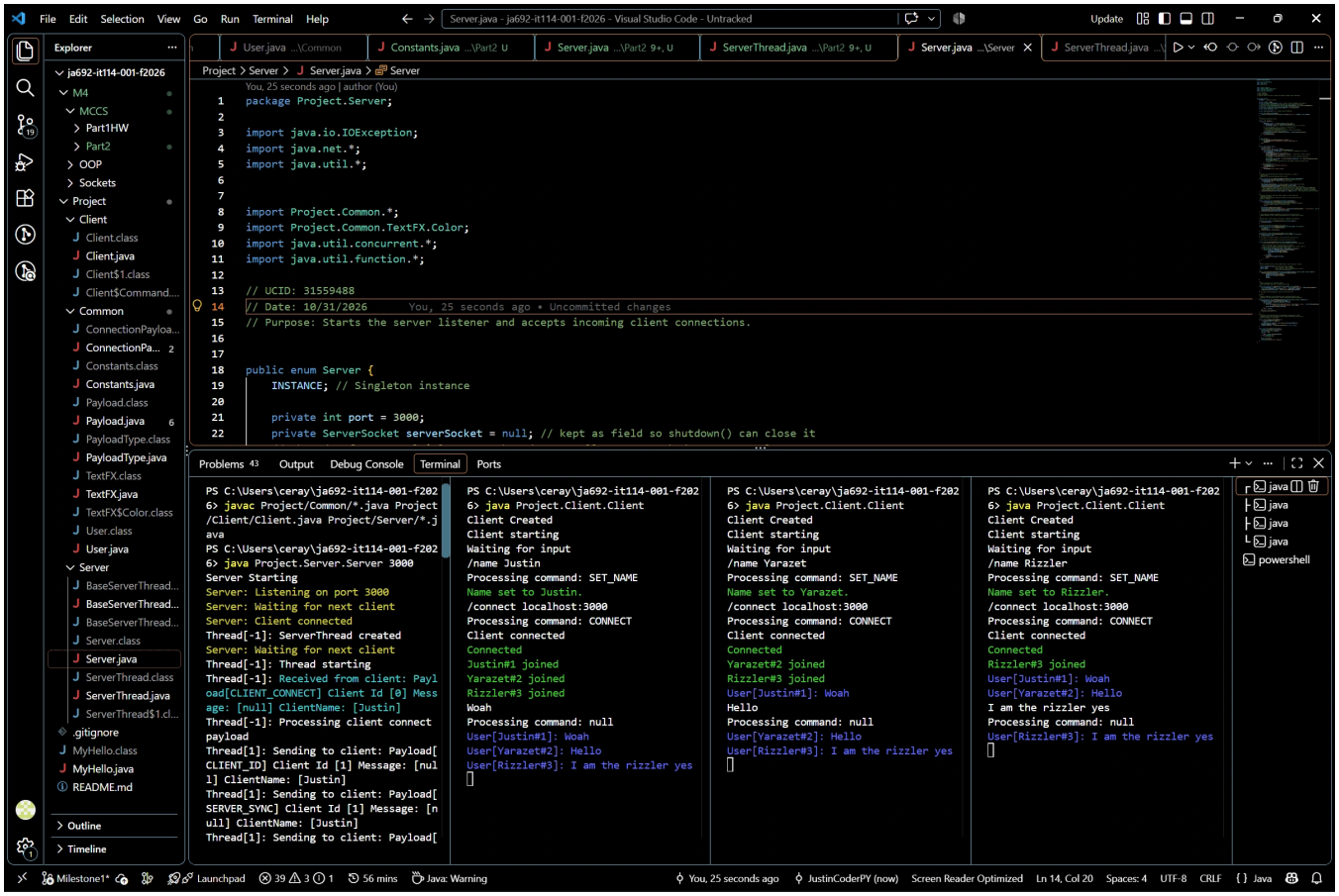
Image evidence prompt

Upload a readable terminal capture showing the server started and listening from the repository root.

Requirements and potential deductions

- The server process visibly starts and remains ready for connections. (up to -0.5 points)

Caption: Server started from the repository root and listened on port 3000



Accept explanation

Response prompt

Explain which method waits for a connection and how your server passes an accepted client to its connection handler.

Requirements and potential deductions

- The response connects the listener, accept call, and handler creation in the submitted source. (up to -0.25 points)

Submitted response

It is the start(int port) method that helps with the ServerSocket with new ServerSocket(port) and then waits for other connection using the .accept(). From there, when the client is connected, the socket is then stored in the incomingClient, passing it to the newServerThread. Lastly, the server used the serverThread.start() so that the client can be handled and the server waits for ne connections.

Multiple clients

Keep at least three clients connected to one server at the same time.

Connection source

URL prompt

Provide direct GitHub file URLs for the server's client registry or callback and the per-client handler on `Milestone1`.

Requirements and potential deductions

- The server keeps independent connections and tracks connected clients. (up to -0.75 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Server/Server.java
https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Server/ServerThread.java

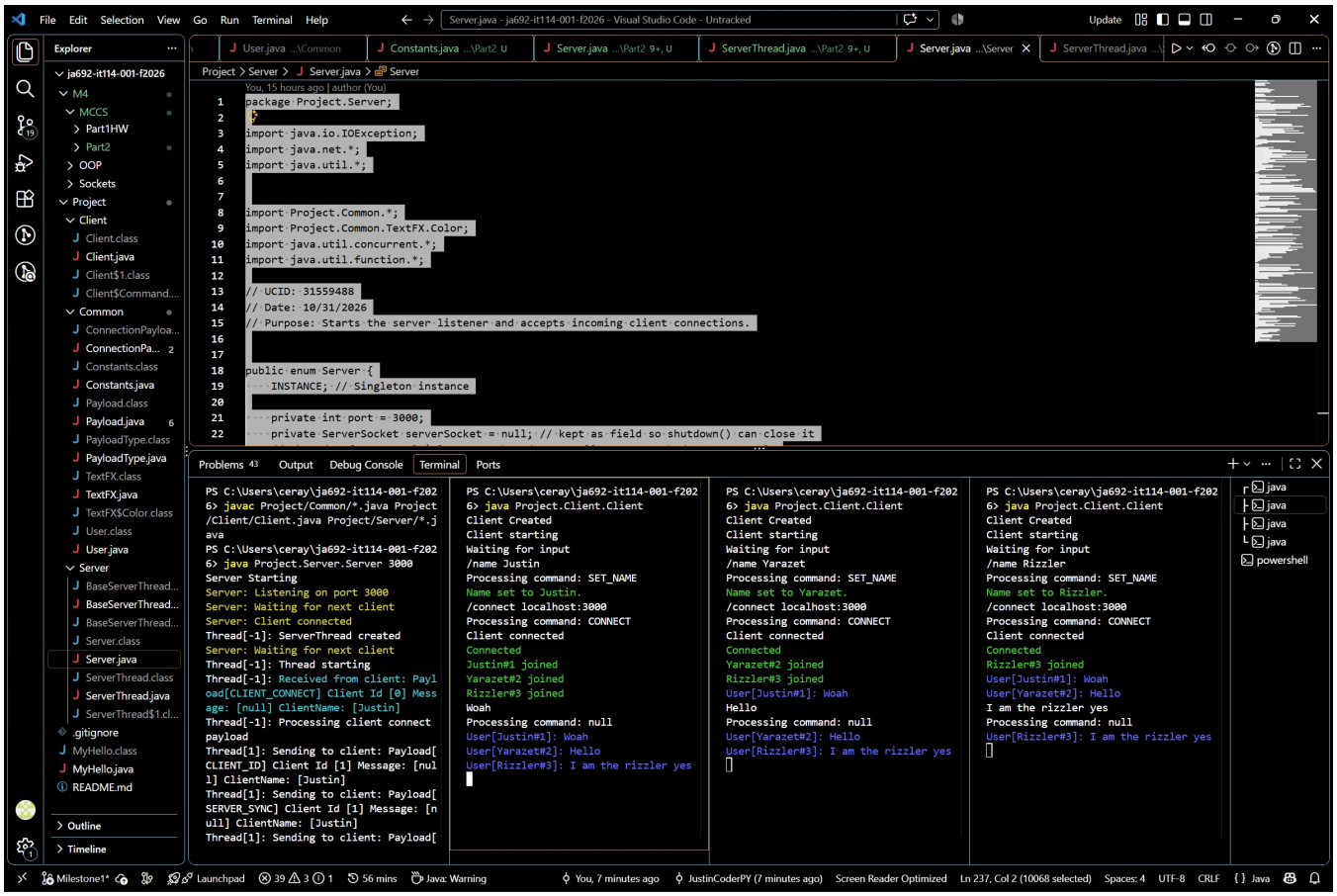
Three-client output

Image evidence prompt

Upload readable terminal captures showing one server and at least three named clients connected simultaneously; split terminals or multiple images are fine.

Requirements and potential deductions

- Three distinct clients are concurrently connected to the same running server. (up to -0.75 points)



Connection explanation

Response prompt

Explain how the submitted server code handles multiple clients without one client's input blocking the others.

Requirements and potential deductions

- The response identifies the actual per-client execution and server tracking in the linked source. (up to -0.5 points)

Submitted response

The server code handles the multiple clients without one client's input blocking the others by using ServerThread so that when user/client is inputting something, it won't block other clients. After that, the server will keep track of the current active clients in connectedClients map, letting more of them connect and talk to each other.

Client identity and connection

Each of three clients sets a name, connects from the command line, and receives connection confirmation.

Client source

URL prompt

Provide direct GitHub file URLs for `/name`, `/connect`, and connection confirmation in the `Project/Client/` code on `Milestone1`.

Requirements and potential deductions

- The client accepts a name before opening a connection and sends that identity to the server. (up to -0.5 points)
- The client processes the server's assigned identity and reports connection confirmation. (up to -0.25 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Client/Client.java

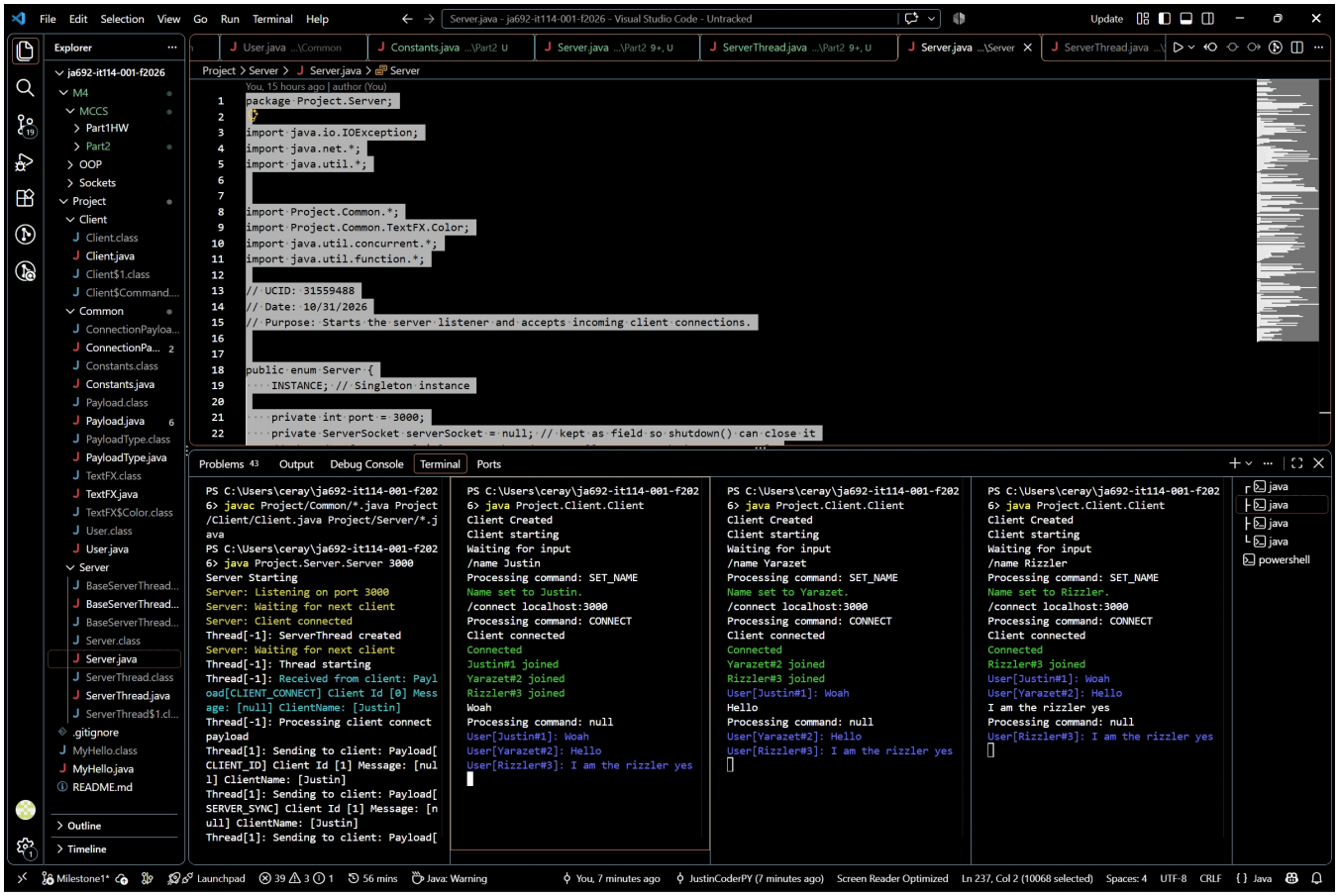
Three-client command output

Image evidence prompt

Upload readable captures of `/name` and `/connect` for each of three clients, including each client's successful connection confirmation.

Requirements and potential deductions

- All three named clients complete `/connect` and show their own confirmation. (up to -0.5 points)



Identity explanation

Response prompt

Explain how your `/name` and `/connect` input reaches the server and how the assigned ID returns to the client.

Requirements and potential deductions

- The response connects command processing, identity sending, server assignment, and client confirmation. (up to -0.25 points)

Submitted response

When using the /name command, it helps save the client's name before the /connect opens to the connection. Once connected

server will ultimately save that information, on top of assigning a unique client ID, and send it back, confirming that the connect valid.

Message exchange

Each connected client can send ordinary chat text that reaches the other clients.

Message source

URL prompt

Provide direct GitHub file URLs for client sending and the server's message routing on `Milestone1`.

Requirements and potential deductions

- The client sends typed message data and the server distributes it to connected clients. (up to -0.75 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Client/Client.java
https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Server/Server.java

Message output

Image evidence prompt

Upload readable captures of a different ordinary message sent by each of the three clients, with the received messages visible on the other clients.

Requirements and potential deductions

- Each of the three clients sends a message received by the other connected clients. (up to -0.75 points)

Caption: Each of the three clients sent a different message, and the messages were received by the other connected clients.



Message explanation

Response prompt

Trace one submitted message from client send through server routing to a receiving client.

Requirements and potential deductions

- The response names the actual send, process, and receive steps in the linked code. (up to -0.5 points)

Submitted response

When clients send a message in the chat, Client.java overall helps create a payload and then sends that to the server. Next, th ServerThread receives and processes that payload, then calls the handleMessage() method. After that, the server used the broadcast() to the formatted message to all connected clients, showing where the client receives and displays it.

Disconnect and reconnect

A client can leave without stopping the server; after server shutdown and restart, clients can reconnect.

Cleanup source

URL prompt

Provide direct GitHub file URLs for client disconnect handling and server connection cleanup on `Milestone1`.

Requirements and potential deductions

- Client and server cleanup close the affected connection and remove its tracked identity without ending unrelated clients. (up to -0.75 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Client/Client.java
https://github.com/JustinCoderPY/ja692-it114-001-f2026/blob/Milestone1/Project/Server/Server.java

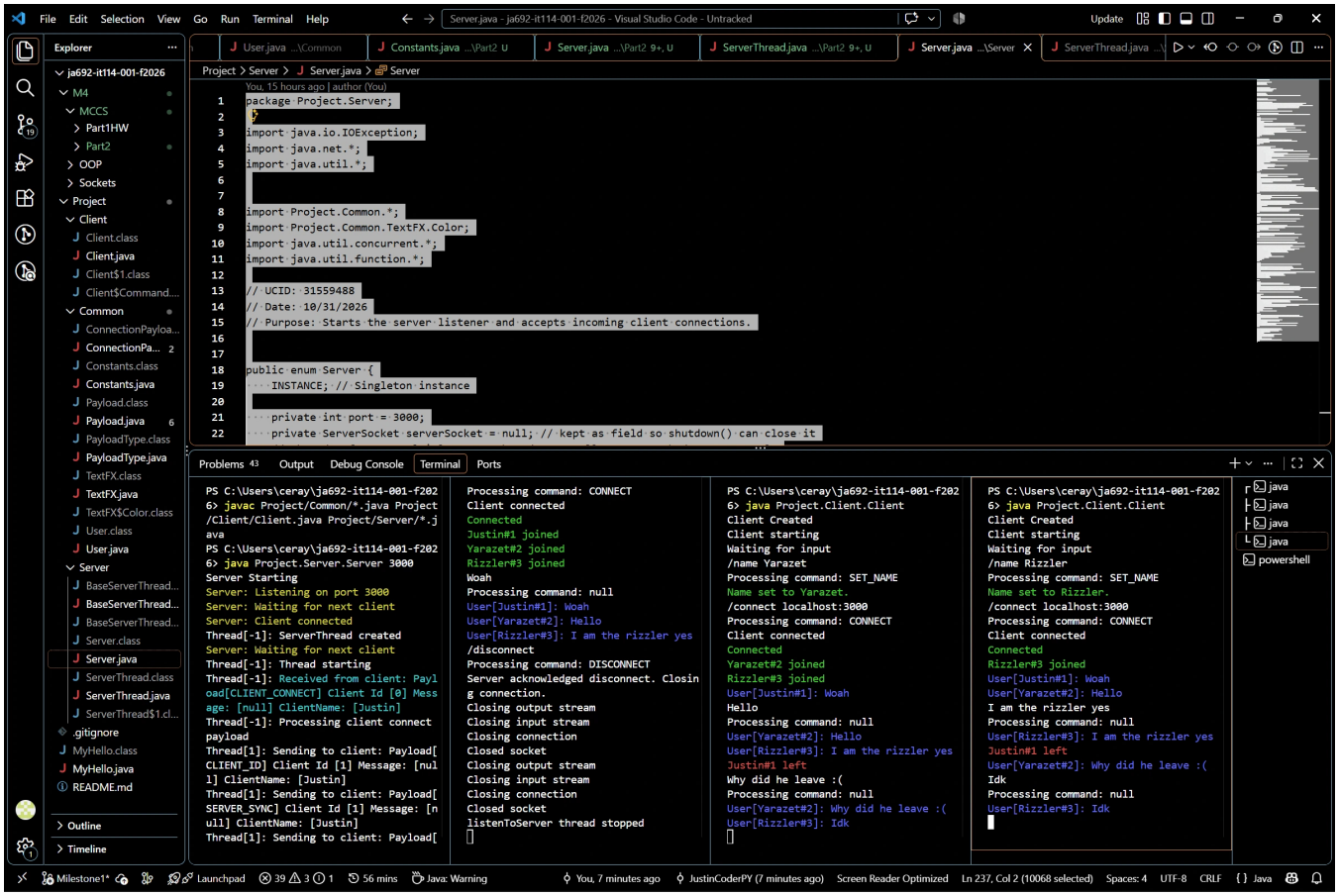
Reconnection output

Image evidence prompt

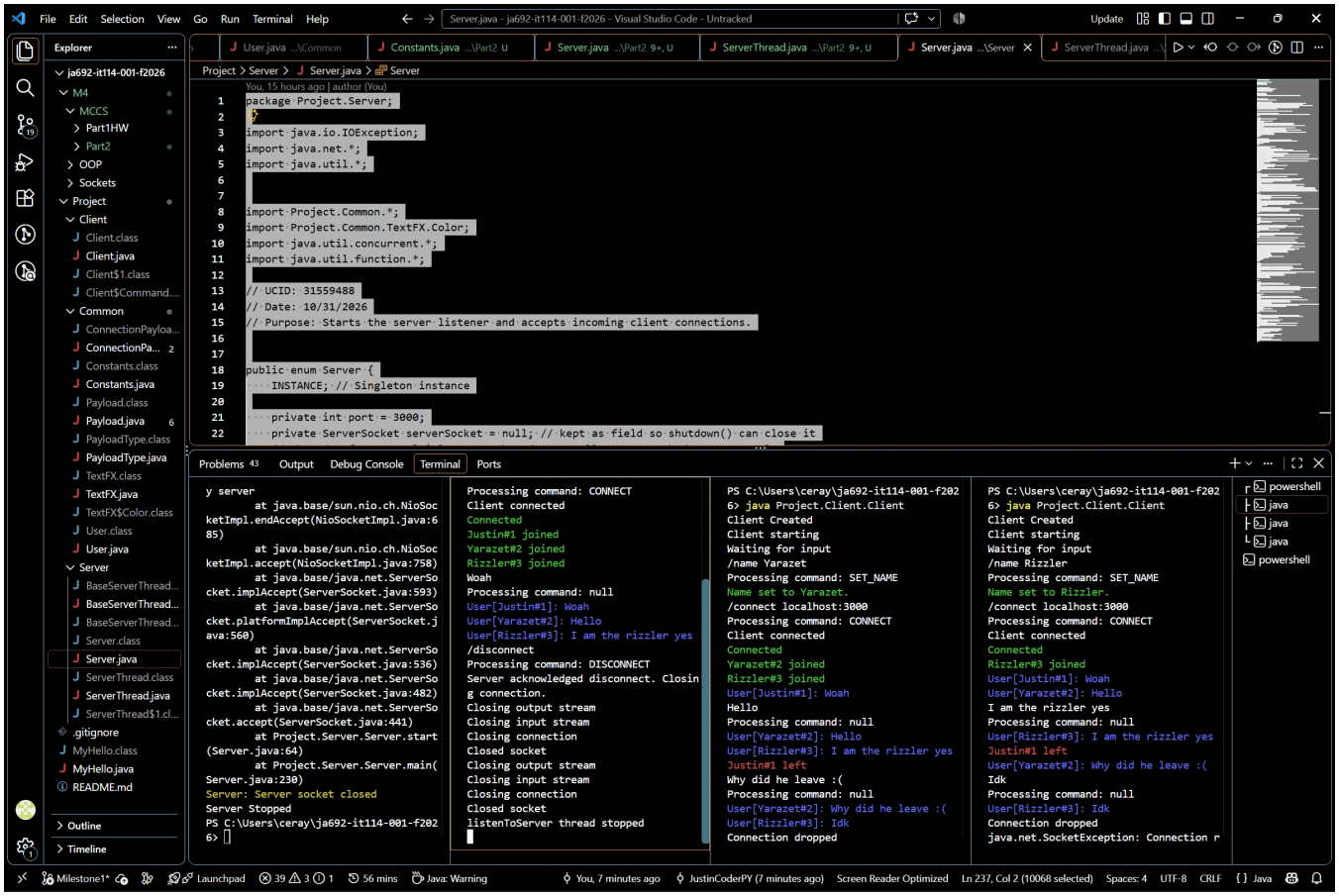
Upload readable terminal captures showing one client disconnect while the server and another client remain active, server shutdown while clients remain open but disconnected, and a successful client reconnect after server restart.

Requirements and potential deductions

- One client leaves while another keeps using the running server. (up to -0.25 points)
- Server shutdown leaves clients running but disconnected. (up to -0.25 points)
- A client reconnects after server restart. (up to -0.25 points)


File Edit Selection View Go Run Terminal Help
Server.java - ja692-it114-001-f2026 - Visual Studio Code - Untracked
Update
Explorer
ja692-it114-001-f2026
M4
MCCS
Part1HW
Part2
OOP
Sockets
Project
Client
Client.class
Client.java
Client$1.class
Client$Command....
Common
ConnectionPayloa...
ConnectionPa... 2
Constants.class
Constants.java
Payload.class
Payload.java 6
PayloadType.class
PayloadType.java
TextFX.class
TextFX.java
TextFX$Color.class
User.class
User.java
Server
BaseServerThread...
BaseServerThread...
BaseServerThread...
Server.class
Server.java
ServerThread.class
ServerThread.java
ServerThread$1.cl...
.gitignore
MyHello.class
MyHello.java
README.md
Outline
Timeline
User.java ...\Common
Constants.java ...\Part2 U
Server.java ...\Part2 9+, U
ServerThread.java ...\Part2 9+, U
Server.java ...\Server
ServerThread.java ...\
Project > Server > Server.java > Server
You, 15 hours ago | author (You)
1 package Project.Server;
2 }
3 import java.io.IOException;
4 import java.net.*;
5 import java.util.*;
6
7
8 import Project.Common.*;
9 import Project.Common.TextFX.Color;
10 import java.util.concurrent.*;
11 import java.util.function.*;
12
13 // UCID: 31559488
14 // Date: 10/31/2026
15 // Purpose: Starts the server listener and accepts incoming client connections.
16
17
18 public enum Server {
19 INSTANCE; // Singleton instance
20
21 private int port = 3000;
22 private ServerSocket serverSocket = null; // kept as field so shutdown() can close it
Problems 43
Output
Debug Console
Terminal
Ports
PS C:\Users\ceray\ja692-it114-001-f2026> javac Project/Common/*.java Project/Client/Client.java Project/Server/*.java
PS C:\Users\ceray\ja692-it114-001-f2026> java Project.Server.Server 3000
Server Starting
Server: Listening on port 3000
Server: Waiting for next client
Server: Client connected
Thread[-1]: ServerThread created
Server: Waiting for next client
Thread[-1]: Thread starting
Thread[-1]: Received from client: Payload[CLIENT_CONNECT] Client Id [0] Message: [null] ClientName: [Justin]
Thread[-1]: Processing client connect payload
Thread[1]: Sending to client: Payload[CLIENT_ID] Client Id [1] Message: [null] ClientName: [Justin]
Thread[1]: Sending to client: Payload[SERVER_SYNC] Client Id [1] Message: [null] ClientName: [Justin]
Thread[1]: Sending to client: Payload[
Processing command: CONNECT
Client connected
Connected
Justin#1 joined
Yarazet#2 joined
Rizzler#3 joined
Woah
Processing command: null
User[Justin#1]: Woah
User[Yarazet#2]: Hello
User[Rizzler#3]: I am the rizzler yes
/disconnect
Processing command: DISCONNECT
Server acknowledged disconnect. Closing connection.
Closing output stream
Closing input stream
Closing connection
Closed socket
Closing output stream
Closing input stream
Closing connection
Closed socket
listenToServer thread stopped
PS C:\Users\ceray\ja692-it114-001-f2026> java Project.Client.Client
Client Created
Client starting
Waiting for input
/name Yarazet
Processing command: SET_NAME
Name set to Yarazet.
/connect localhost:3000
Processing command: CONNECT
Client connected
Connected
Yarazet#2 joined
Rizzler#3 joined
User[Justin#1]: Woah
Hello
Processing command: null
User[Yarazet#2]: Hello
User[Rizzler#3]: I am the rizzler yes
Justin#1 left
Why did he leave :(
Processing command: null
User[Yarazet#2]: Why did he leave :(
User[Rizzler#3]: Idk
PS C:\Users\ceray\ja692-it114-001-f2026> java Project.Client.Client
Client Created
Client starting
Waiting for input
/name Rizzler
Processing command: SET_NAME
Name set to Rizzler.
/connect localhost:3000
Processing command: CONNECT
Client connected
Connected
Rizzler#3 joined
User[Justin#1]: Woah
User[Yarazet#2]: Hello
I am the rizzler yes
Processing command: null
User[Rizzler#3]: I am the rizzler yes
Justin#1 left
User[Yarazet#2]: Why did he leave :(
Idk
Processing command: null
User[Rizzler#3]: Idk
java
java
java
java
powershell
Milestone1*
Launchpad
39 3 1
56 mins
Java: Warning
You, 7 minutes ago
JustinCoderPY (7 minutes ago)
Screen Reader Optimized
Ln 237, Col 2 (10068 selected)
Spaces: 4
UTF-8
CRLF
Java

Caption: The server was shut down while the client terminals remained open and detected that the connection was dropped.



Caption: After the server was restarted, the client reconnected successfully and showed the Connected confirmation.



Cleanup explanation

Response prompt

Explain which client and server cleanup steps permit disconnect and reconnect without a stale client identity.

Requirements and potential deductions

- The response connects actual socket closing, registry removal, and new connection setup. (up to -0.5 points)

Submitted response

Whenever a client gets disconnected, both the streams and sockets also close down with it, removing the client from connectedClients. However, when the client reconnects again, a fresh new socket and ServerThread will be created, assigning to a new client ID, preventing the old connection and identity from remaining active.

Reflection

Pull request URL

URL prompt

Submit the pull-request URL from `Milestone1` into `main`.

Requirements and potential deductions

- The pull request contains the Milestone 1 source and remains open for the exported PDF commit. (up to -0.2 points)

https://github.com/JustinCoderPY/ja692-it114-001-f2026/pull/9

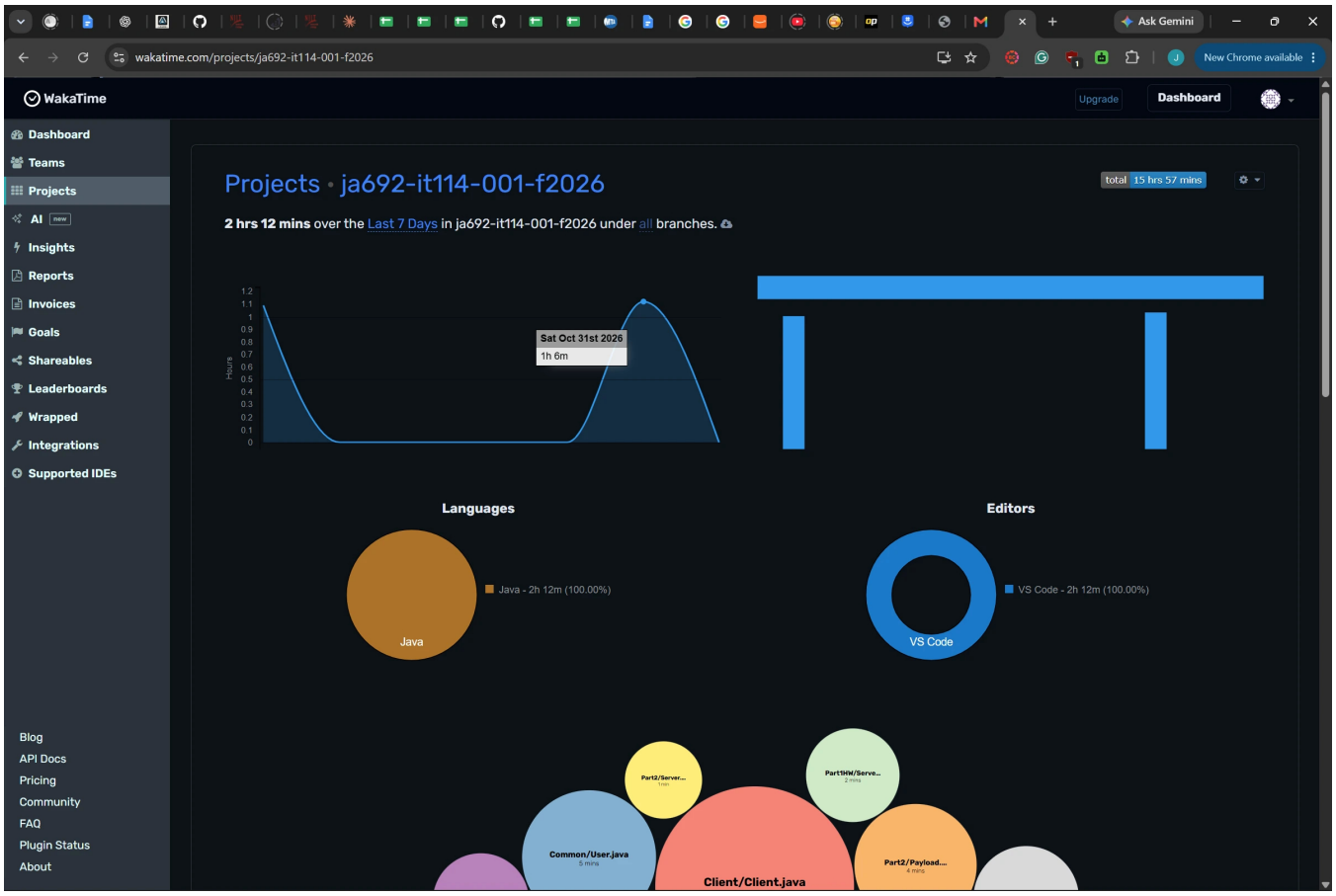
WakaTime project overview

Image evidence prompt

Upload a readable WakaTime project overview showing your repository name; time duration is not graded.

Requirements and potential deductions

- The capture identifies the project repository. (up to -0.1 points)



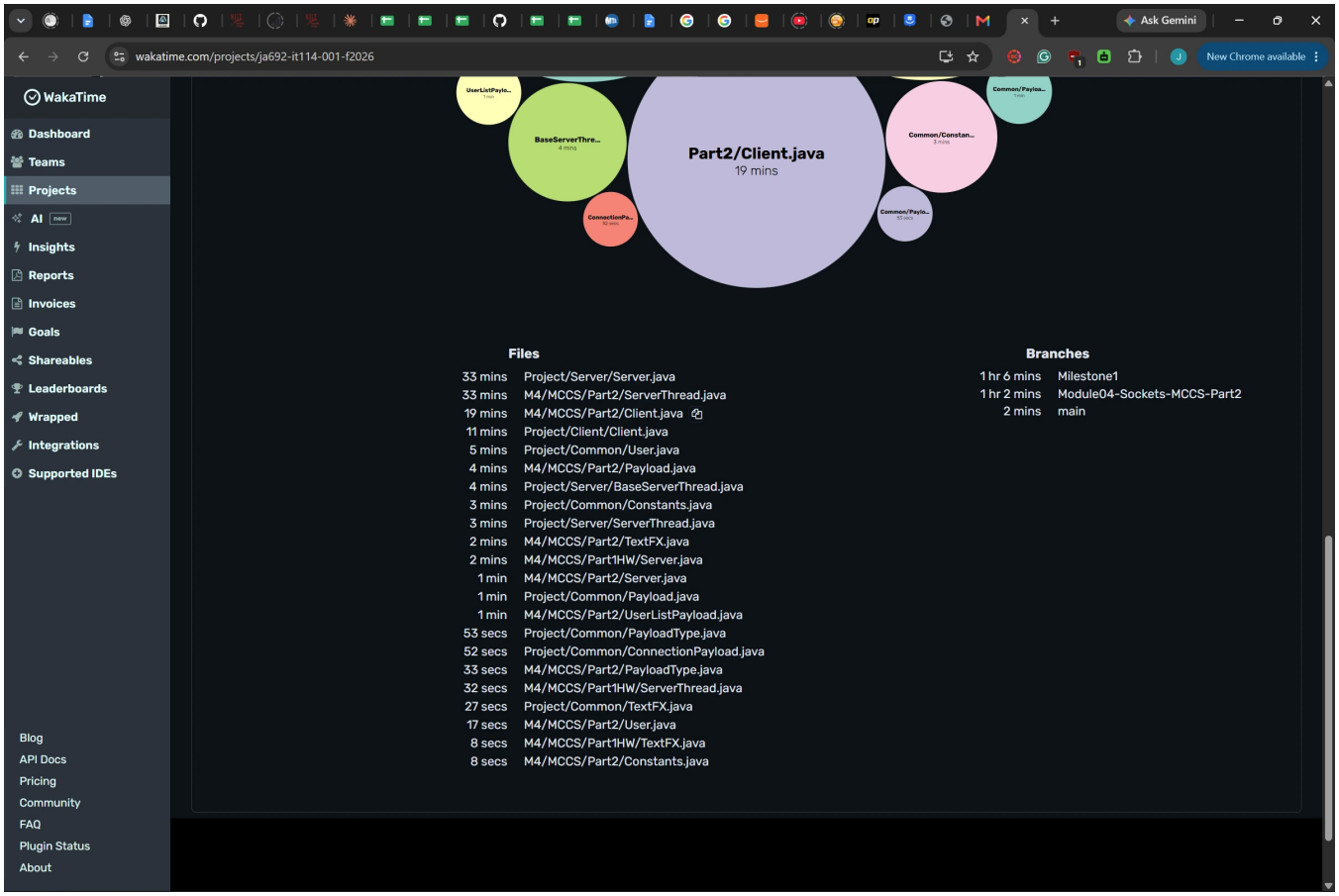
WakaTime file activity

Image evidence prompt

Upload a readable WakaTime file-activity capture for this milestone; time duration is not graded.

Requirements and potential deductions

- The capture identifies relevant Java file activity. (up to -0.1 points)



Learned or verified

Response prompt

What did you learn or verify about the project framework and its connection behavior?

Requirements and potential deductions

- The response identifies a concrete lesson or verification result. (up to -0.2 points)

Submitted response

I basically learned what I need to prepare for this project. Since the selection I'll be picking is a battleship, I can see that the clie server will need to constantly communicate so both players can be updated with the correct game information.

Easiest part

Response prompt

Which part of this milestone was easiest, and why?

Requirements and potential deductions

- The response names a specific part and explains why it was easier. (up to -0.2 points)

Submitted response

The easiest part of the milestone was understanding how the client connection worked. Once I saw how each client needed a r and connection to the server, with an output of a confirmation, the flow becomes a lot easier to know.

Difficult or uncertain part

Response prompt

Which part was most difficult or remains uncertain, and what did you try or check?

Requirements and potential deductions

- The response names a specific difficulty or uncertainty and a relevant check. (up to -0.2 points)

Submitted response

The challenging aspect of this assignment was more so the server aspect. I needed to know how servers were able to accept c
and why a new thread had to be created rather than saved, and how it even keeps track of it to begin with.